

Team Number: 011-3

Team Name: CU Swap

Group members names, emails, and github usernames:

Sashi Wolf

Email: sawo7759@colorado.edu

GitHub username: sashewolf

Hayden Leovy

Email: hale1973@colorado.edu

GitHub username: HaydenLeovy

Shon Butz

Email: shbu4101@colorado.edu

GitHub Username: ShonButz

Dylon Gregory

Email: dygr1357@colorado.edu

GitHub username: Dtate21

Joe Young

Email: No email yet because Joe wasn't in lab or answering the phone

GitHub: JYJY-UCB

Application Name: CU Swap

Application Description:

CU Swap is a marketplace app designed exclusively for University of Colorado Boulder students. It provides a safe, easy-to-use platform where students can buy, sell, and trade items within the CU community. Users can upload photos, write detailed descriptions, and set prices for their listings, while buyers can browse categories and arrange on-campus meetups. Access to the app is restricted to CU students through verified university email addresses, ensuring a trusted environment.

The main value of CU Swap lies in its focus on convenience and community. Unlike large-scale platforms such as Facebook Marketplace, CU Swap operates on a smaller, student-centered scale. This makes transactions faster, communication simpler, and listings more relevant to campus life. To maintain quality and safety, the app will include moderator accounts with administrative privileges to oversee listings.

Audience:

CU students looking to buy or sell items. Ideal users are someone who is looking to quickly and easily sell or buy items that are relevant for CU students. Our software aims to facilitate this in a quick and convenient way.

Vision Statement:

For CU students, who want to trade or sell items. The CU swap is a marketplace that allows CU students specifically to buy, sell, and trade among the CU community. Unlike Amazon, our product is focused specifically on items that are appealing to CU students.

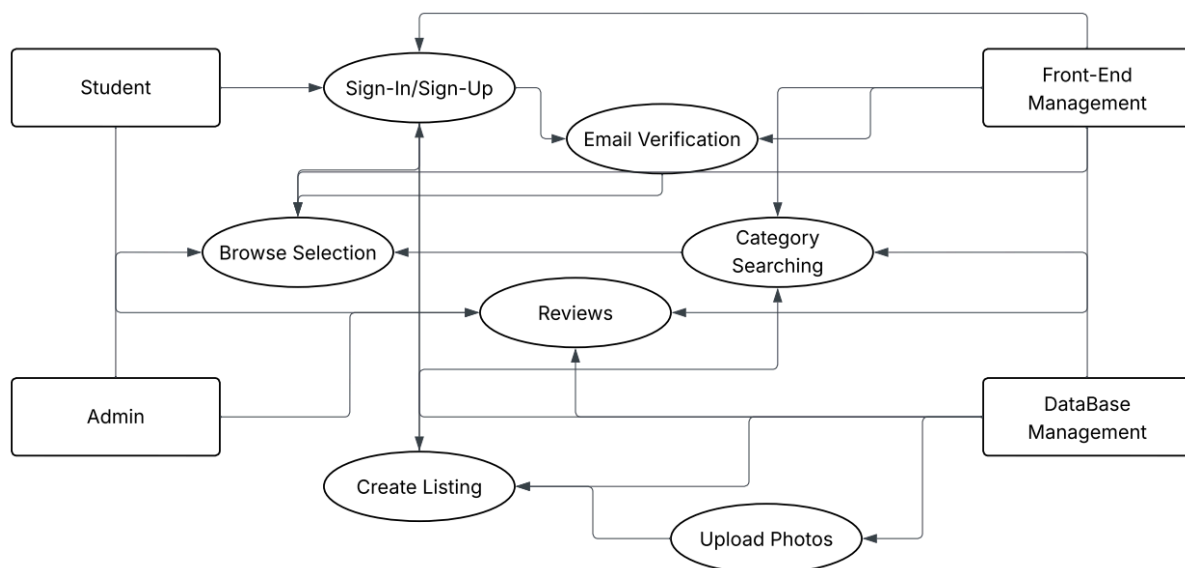
Development Methodology: Agile Methodology

Communication Plan: Our plan is to use Microsoft Teams as our main method of communication. We will also have a text group chat as a backup.

Team Meeting:

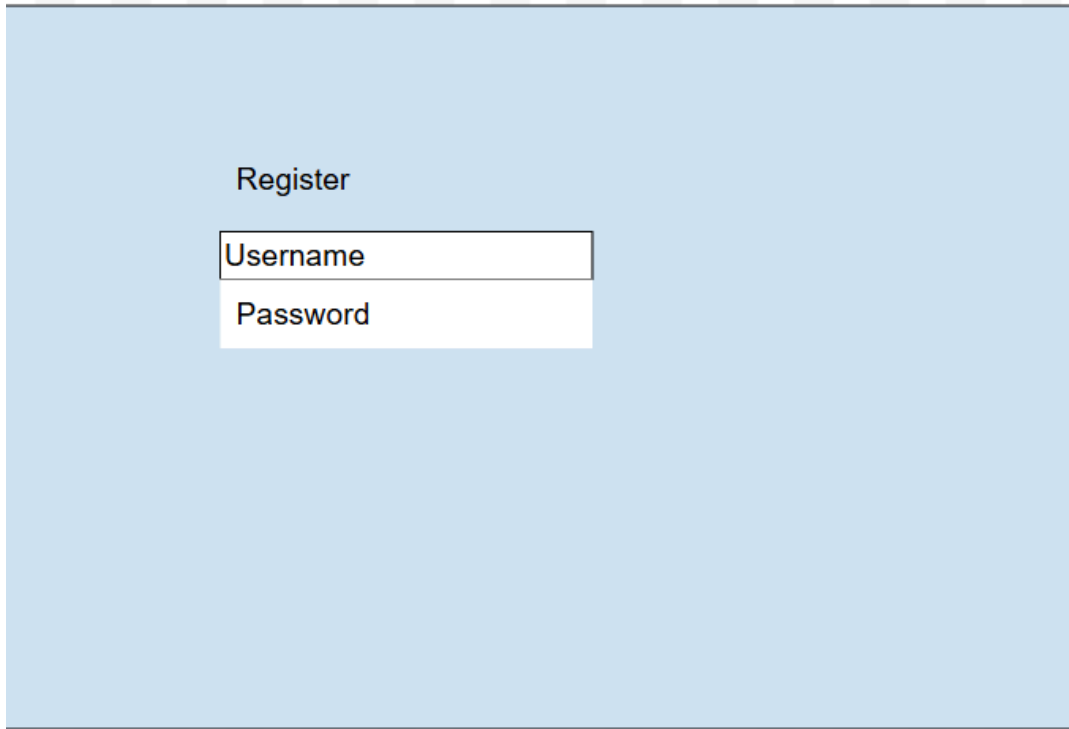
Weekly Meeting (TA): 4:45 PM Tuesday

Team Meetings: 11:15 AM Thursday (In-Person)

Use Case Diagram

Wireframe

Register page

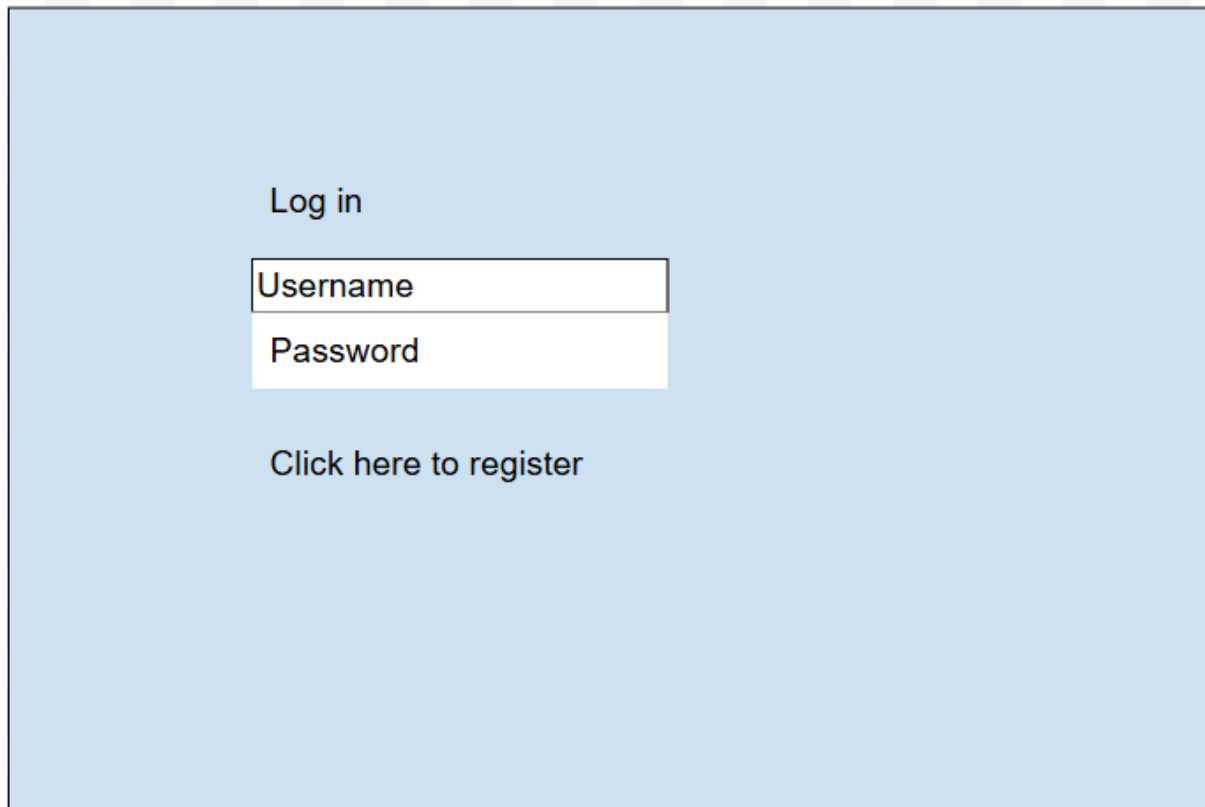
A wireframe of a register page. It features a light blue background. In the upper left quadrant, the word "Register" is centered. Below it, there are two stacked white input fields. The top field is labeled "Username" and the bottom field is labeled "Password".

Register

Username

Password

Login page

A wireframe of a login page. It features a light blue background. In the upper left quadrant, the words "Log in" are centered. Below them, there are two stacked white input fields. The top field is labeled "Username" and the bottom field is labeled "Password". Below the input fields, the text "Click here to register" is centered, serving as a link to the registration page.

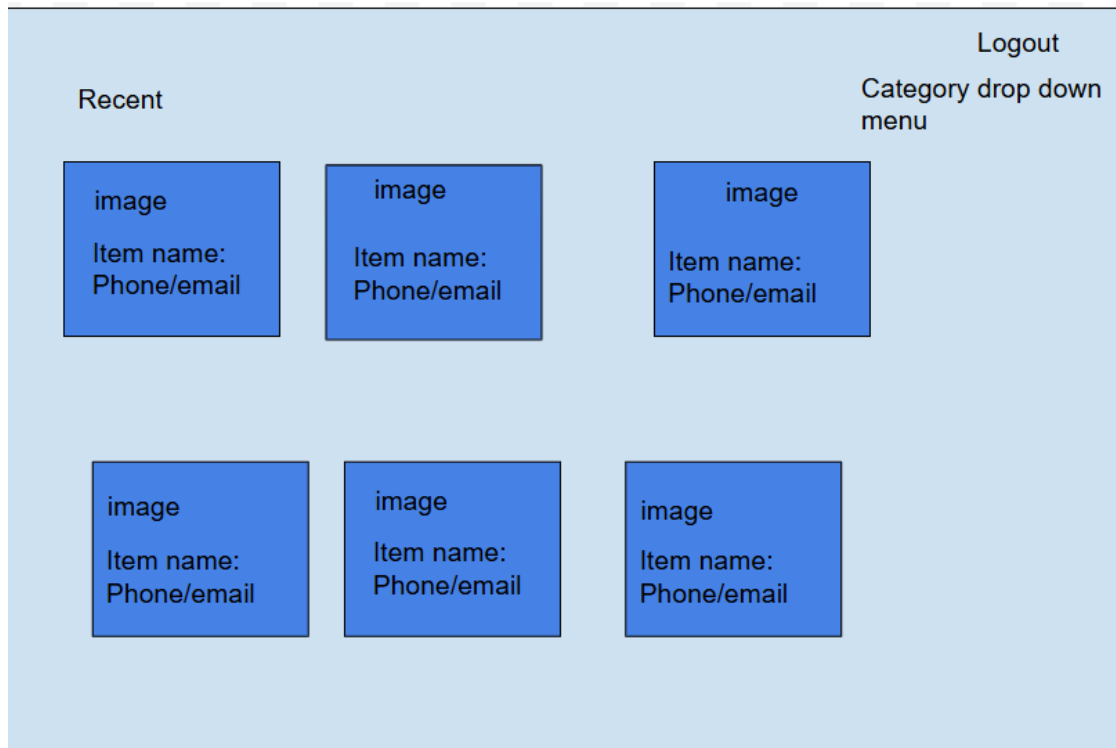
Log in

Username

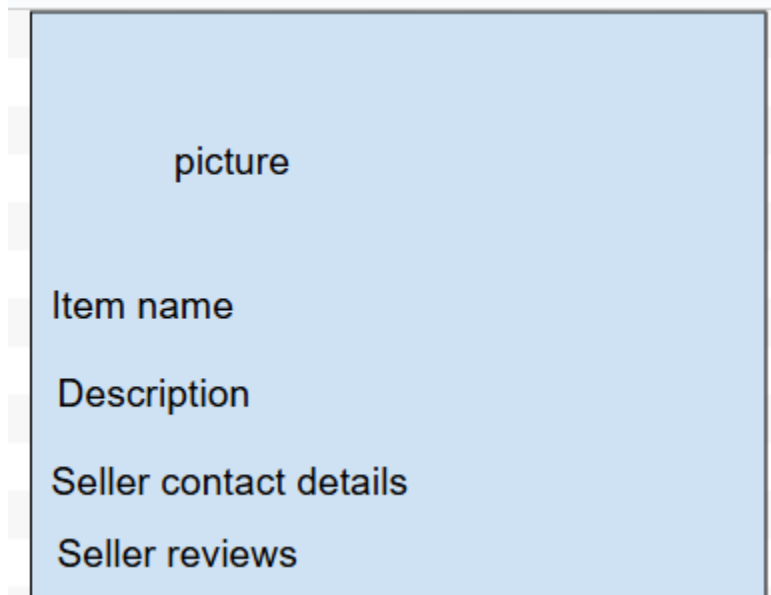
Password

[Click here to register](#)

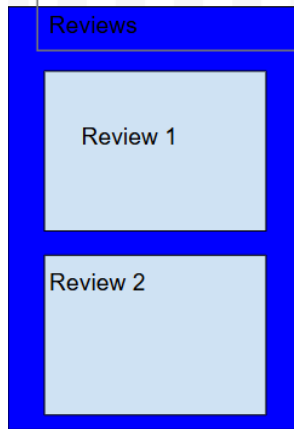
Home page



Clicked on item



Clicked on review



Extra Credit: Potential Risks and Mitigation Strategies

Risk: Inappropriate or fake posts

Severity: High

Fix: Have moderators who can delete bad posts and delete users who break the rules, and not allow them to create a new account with the same email.

Risk: Privacy or data leaks

Severity: High

Fix: Require CU student email sign-in and use an authentication system to keep personal information and messages safe. Hash passwords.

Risk: Code bugs that break main features

Severity: High

Fix: Test each feature as it's built using small test cases. Review code regularly and fix errors before merging new updates.

Risk: Poor database structure or connection issues

Severity: Medium

Fix: Plan the database early and keep it simple. Use clear relationships between users, listings, and messages, and test all database queries.

Risk: Merge conflicts or inconsistent code styles in group work

Severity: Medium

Fix: Use version control tools like GitHub. Set clear coding rules for the group and review pull requests before merging changes.